Data Science : Assignment – 2

Submitted By: Group 14

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1. Introduction

The purpose of this assignment is to acquire a profound understanding of two different machine learning algorithms, classification and Regression. The objective is to evaluate the models, compare their performance, identify the most suitable one for the problem, and subsequently optimize it to enhance its predictive capabilities.

The first task involved a classification problem in which the goal was to predict whether an individual earns more than $50,000 per year using the Adult Census Income dataset. The regression problem focused on predicting housing prices in California.

The assignment involves performing data preprocessing and feature selection, followed by training at least three different machine-learning classifiers/regressors. Their performance is then compared using 10-fold cross-validation based on two evaluation metrics, such as MSE and R^2. After identifying the best-performing classifier/regressor, one of its key parameters is optimized according to the cross-validation results. Finally, the selected classifier/regressor with the optimized parameter is trained on the full dataset, and the resulting predictor model is saved.

KNIME is a single platform for data science applications. It is a user-friendly, open-source platform that enables efficient data integration, processing, supporting many machine learning algorithms and visualizations through its intuitive drag-and-drop workflow interface. To achieve the goal of this assignment, we opted for the KNIME platform, and it produced a good result.

2. Classification

Classification is used to build predictive models that categorize observations into predefined classes, allowing us to make decisions or predictions based on learned patterns from historical data.

2.1 Data Preprocessing and Feature Selection

The dataset is Adult Census Income dataset, which contains irrelevant features for predicting income and missing values in the form”?”. Initially, there are 14 attributes in the dataset. At first, we replace the missing values with the most frequent value of that attribute. After that, perform feature selection. The rank correlation method is used for feature selection because it measures the strength of the relationship between each attribute and the target variable using their ranked values. Identifying features that have

strong positive or negative correlations with the target variable helps to remove irrelevant or redundant attributes. At last, we took most suitable 5 features for classification. It makes the dataset clearer and easier to analyse. The following KNIME workflow depicts the above process.

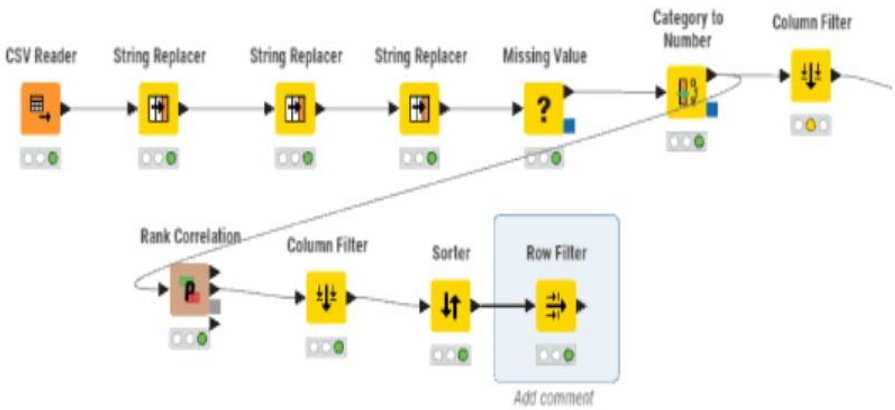

Figure 1: KNIME Workflow – Data Preprocessing and Feature Selection

2.1.1 Exploratory Data Analysis

For analysing the dataset, the KNIME visualization methods are very informative. The following boxplot represents the relationship between age and income. It shows that the median age of individuals earning >50K is higher than those earning <=50K. Moreover, both categories have outliers at higher ages. This indicates that age is an important feature for predicting income, as older individuals are more likely to earn a higher income.

The correlation matrix shows the strength of the relationship between two different features and from which we select the features that have a correlation value greater than or equal to 0.2.

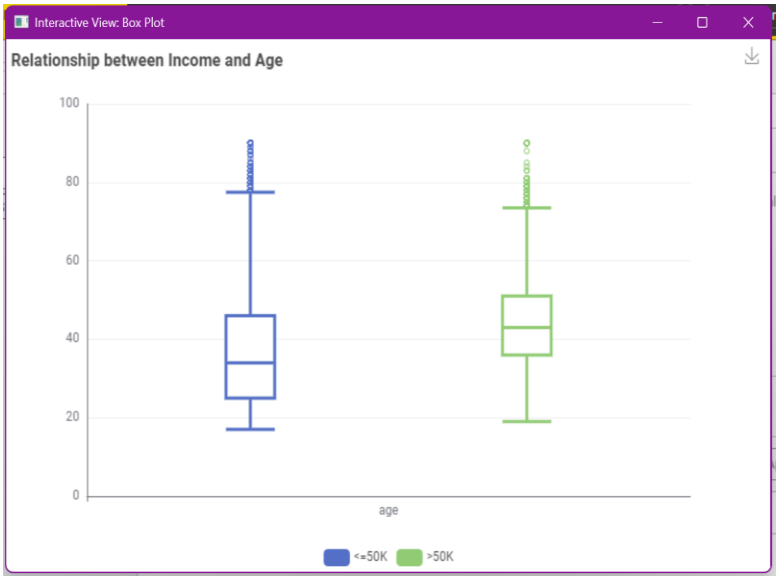

Figure 2: Box Plot-Relationship between Income and Age

Correlation matrix (Table)

Rows: 14 | Columns: 14

#	RowID	age	workclass	fnlwgt	education	educatio...	marital-s...	occupation	relations...	gender	capital-g...
1	age	1	0.187	-0.078	0.089	0.063	0.513	0.004	0.179	-0.1	0.124
2	workclass	0.187	1	-0.051	0.082	0.13	0.067	-0.146	-0.018	-0.05	0.044
3	fnlwgt	-0.078	-0.051	1	-0.003	-0.03	-0.034	0.015	-0.005	-0.028	-0.009
4	education	0.089	0.082	-0.003	1	0.562	-0.005	-0.006	0.025	-0.023	0.076
5	educational-num	0.063	0.13	-0.03	0.562	1	-0.017	0.066	0.018	-0.003	0.119
6	marital-status	0.513	0.067	-0.034	-0.005	-0.017	1	0.006	0.277	0.041	0.055
7	occupation	0.004	-0.146	0.015	-0.006	0.066	0.006	1	-0.021	-0.047	0.02
8	relationship	0.179	-0.018	-0.005	0.025	0.018	0.277	-0.021	1	0.394	-0.006
9	gender	-0.1	-0.05	-0.028	-0.023	-0.003	0.041	-0.047	0.394	1	-0.065
10	capital-gain	0.124	0.044	-0.009	0.076	0.119	0.055	0.02	-0.006	-0.065	1
11	capital-loss	0.058	0.026	-0.001	0.043	0.077	0.025	0.021	-0.008	-0.043	-0.066
12	hours-per-week	0.147	0.034	-0.016	0.087	0.164	0.127	0.1	0.009	-0.264	0.092
13	native-country	-0.007	-0.037	0.073	0.108	-0.054	0.003	-0.048	0.063	-0.01	-0.016
14	income	0.269	0.093	-0.006	0.197	0.328	0.13	0.085	-0.078	-0.215	0.278

Figure 3: Correlation matrix (Table)

Included Rows (Table)

Rows: 5 | Columns: 1

#	RowID	income Number (Float)
1	hours-per-week	0.268
2	age	0.269
3	capital-gain	0.278
4	educational-num	0.328
5	income	1

Figure 4: Included Rows (Table)

2.2 Machine Learning Classifiers: Selection and Comparison

2.2.1 Selection

For this classification problem, we adopt three classification algorithms.

1. Logistic Regression

Logistic Regression is a supervised classification algorithm used when the target variable is categorical. We aim to classify the income into two categories. This method has many strengths, such as being simple and interpretable, working well on linearly separable features, fast to train, and supporting regularisation to avoid overfitting.

2. Random Forest

   It is one of the most powerful and commonly used machine-learning models. A Random Forest is a supervised ensemble classification algorithm that combines many decision trees to make more accurate and stable predictions. While our dataset has many features, it is very useful. The main advantages of this method are handles non-linear relationships, robust to missing data, less sensitive to outliers, performs feature importance ranking, works well without heavy preprocessing.

3. K Nearest Neighbour(K=3)

   KNN is a supervised machine-learning algorithm used for classification. It is a distance-based, instance-based, non-parametric algorithm. It makes predictions based on the closest data points in the training set. It is simple and intuitive, no need of training time, effective for non-linear boundaries, performs well with small datasets.

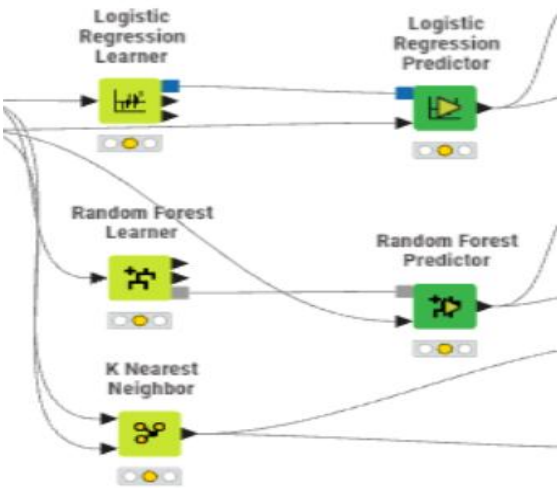

Figure 5: Machine Learning Classifiers

2.2.2 Comparison

For comparing the different classifiers 10-fold cross validation method is used. It evaluated the performance of machine learning models and making the efficient use of dataset. This method splits the dataset into 10 equal parts (folds). The model is trained and evaluated 10 times, each time using a different fold as the test set and the remaining 9 folds as the training set.

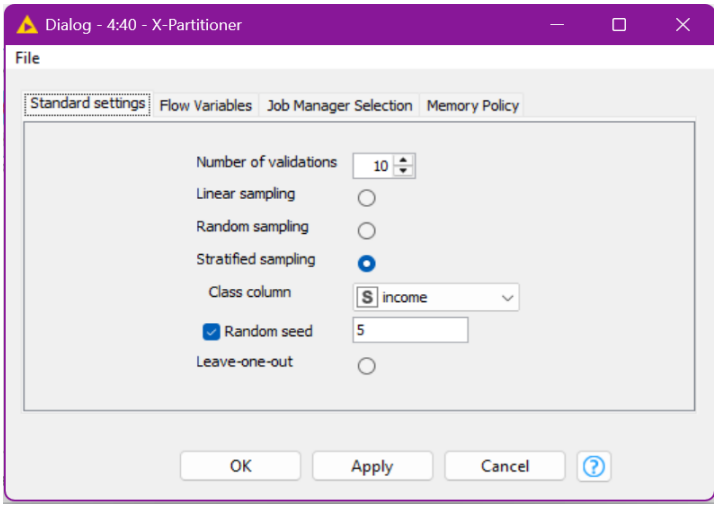

Figure 6: 10-fold cross validation method

For comparison, two dominating measures, Accuracy and Area Under Curve (AUC) are used since these performance metrics are best in capturing the classification capacity of the model and clearly differentiate between classes. From the table below it is clear that Random Forest Classifier is more accurate compared to the others and the AUC is also high for it.

Model	**Accuracy (%)**	**AUC**
Logistic Regression	81.6%	Moderate
Random Forest	**83.2%**	**High**
KNN	80.5%	Low–Moderate

Figure 7: Comparison Table

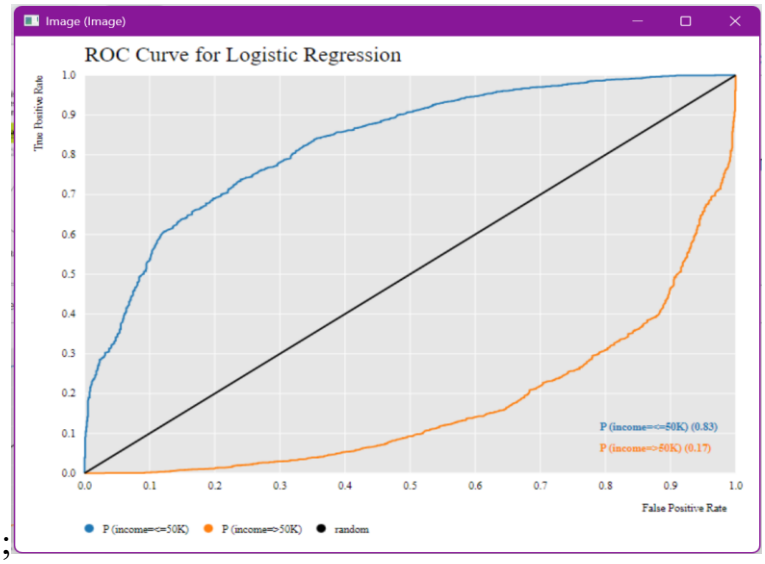

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Figure 8: ROC curve for Logistic Regression

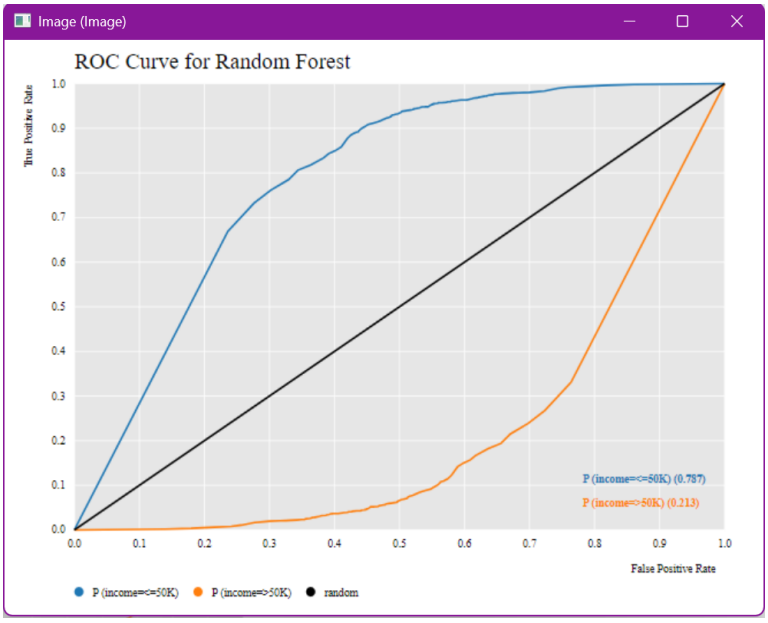

Figure 9: ROC curve for Random Forest

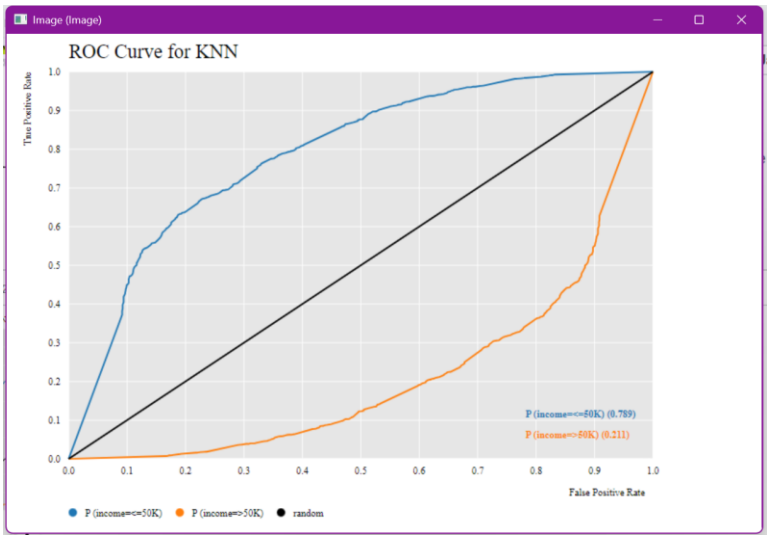

Figure 10: ROC curve for KNN

2.3 Model Optimization

It is an important step in machine learning task as it can enhance the performance of a model. We have used Parameter optimization Loop to get a structured approach to explore various combinations of parameters. This will help to identify which optimal setting can improve the accuracy of a model. We have used Parameter optimization only for the Random Forest model as it gave us better model when compared to other models.

Random Forest Learner allow us to change many parameters in KNIME like the max number of levels, no of models, node size, child size etc. We have changed several

parameters and have not observed much difference in the performance of the model, but changing the tree depth had changed improved the performance of metrics to a certain extent and got new accuracy to 86.9%. Hence, we can say that the tree depth in a random forest model is a crucial hyperparameter that influences the trade-off between model complexity and generalization. With the help of looping concept in KNIME we have trained the model and obtained the best parameter with better accuracy than before.

Tree Depth	Accuracy	AUC
15	82.8	78
25	83.2	78.7
30	83.2	78.7

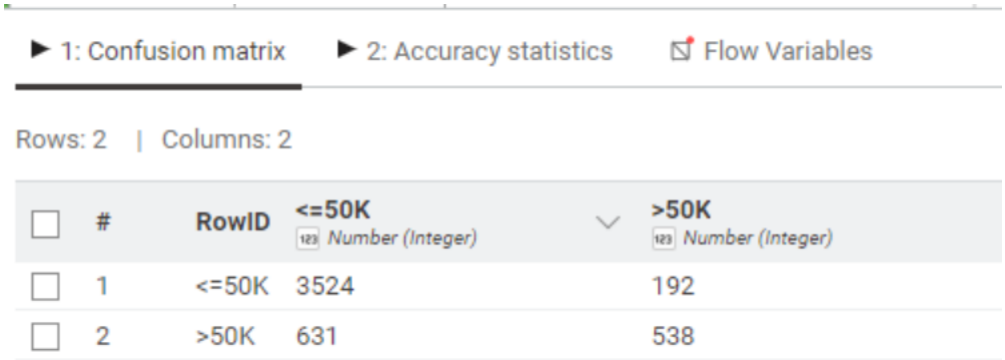

Figure 10: *Confusion Matrix after tuning*

This proves tree depth is a decisive hyperparameter, controlling the balance between overfitting and generalization

2.4 Train and Save the Model

After the best model (Random Forest Classifier) is optimized, it is then trained on the whole dataset which will make it more optimized and then this final model is saved. The Model Writer node in KNIME is used to store a trained model so it can be reused or deployed later. The saved model can then be loaded using the Model Reader node in another workflow.

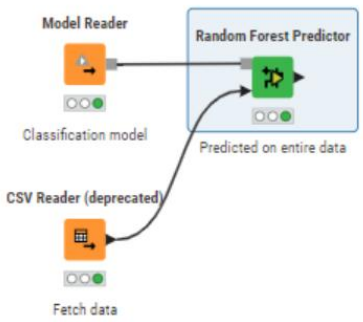

Figure 10: Prediction Workflow

Rows: 48842 | Columns: 8

#	RowID	age (Number (Float))	workclass (String)	marital-status (String)	occupation (String)	hours-per-week (Number (Float))	native-country (String)	income (String)	Prediction (inc... (String)
1	Row0	25	Private	Never-married	Machine-op-inspct	40	United-States	<=50K	<=50K
2	Row1	38	Private	Married-civ-spouse	Farming-fishing	50	United-States	<=50K	<=50K
3	Row2	28	Local-gov	Married-civ-spouse	Protective-serv	40	United-States	>50K	<=50K
4	Row3	44	Private	Married-civ-spouse	Machine-op-inspct	40	United-States	>50K	>50K
5	Row4	18		Never-married		30	United-States	<=50K	<=50K
6	Row5	34	Private	Never-married	Other-service	30	United-States	<=50K	<=50K
7	Row6	29		Never-married		40	United-States	<=50K	<=50K
8	Row7	63	Self-emp-not-inc	Married-civ-spouse	Prof-specialty	32	United-States	>50K	>50K
9	Row8	24	Private	Never-married	Other-service	40	United-States	<=50K	<=50K
10	Row9	55	Private	Married-civ-spouse	Craft-repair	10	United-States	<=50K	<=50K
11	Row10	65	Private	Married-civ-spouse	Machine-op-inspct	40	United-States	>50K	>50K
12	Row11	36	Federal-gov	Married-civ-spouse	Adm-clerical	40	United-States	<=50K	>50K
13	Row12	26	Private	Never-married	Adm-clerical	39	United-States	<=50K	<=50K
14	Row13	58		Married-civ-spouse		35	United-States	<=50K	<=50K
15	Row14	48	Private	Married-civ-spouse	Machine-op-inspct	48	United-States	>50K	>50K

Figure 11: Table with prediction for entire dataset

3. Regression

Regression is a type of supervised machine-learning technique used to predict continuous numerical values. The goal of this assignment is to train a regression model that predicts the medianHouseValue, which represents the median dollar value of houses within each housing block.

3.1 Data Preprocessing and Feature Selection

The dataset contains irrelevant features for predicting medianHouseValue and missing values in the form"?". Initially, there are 10 attributes in the dataset. At first, we replace the missing values with the mean value of that attribute. After that, we created three brand new features that seemed more meaningful than just the raw numbers:

1. rooms per household - how many rooms per house on average (got this by dividing total rooms by number of households)

2. bedrooms per room - what fraction of the rooms are bedrooms

3. population per household - tells you how crowded the neighbourhood is

Turns out these new features helped quite a bit because they captured relationships that weren't obvious in the original data. It makes the dataset clearer and easier to analyse. The following KNIME workflow depicts the above process.

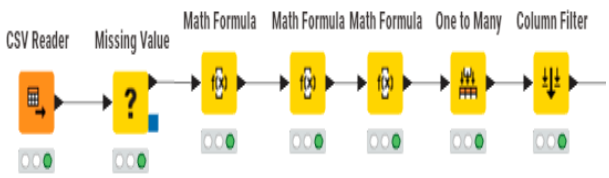

Figure 12: Workflow for Feature Selection

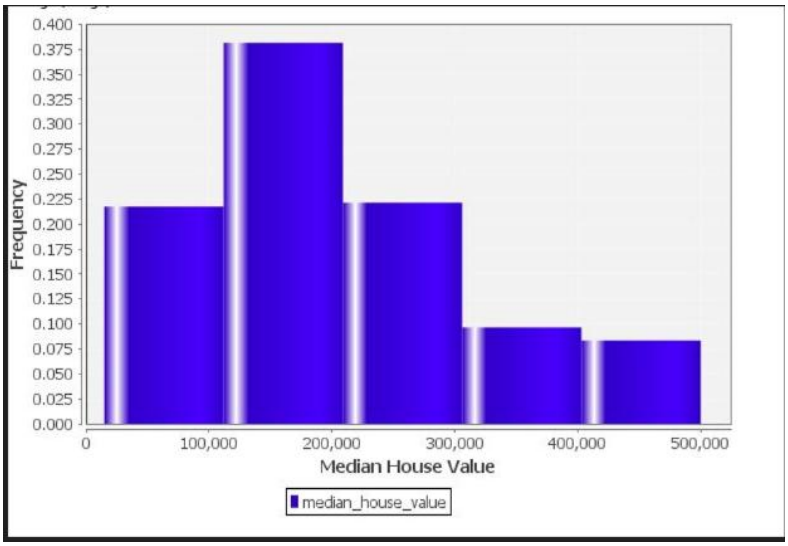

Figure 13: Distribution of median house values

The histogram shows the distribution of median house values across California districts. Most homes fall within the $100,000–$250,000 range, indicating that most neighbourhoods in the dataset have moderate housing prices. There is a noticeable decline in frequency as values increase, showing that high-value homes above $300,000 are far less common. The sharp accumulation near $500,000 reflects the dataset's upper limit, meaning many expensive districts hit the capped maximum value rather than representing true market prices.

3.2 Machine Learning Regressors: Selection and Comparison

3.2.1 Selection

For this regression problem, we adopt three regression algorithms.

1.Tree Ensemble Leaner

The Tree Ensemble Learner is a machine-learning algorithm that builds multiple decision trees and combines their outputs to produce a more accurate and stable prediction. This makes it especially effective for predicting the median house value, where numerous interacting factors—such as income, housing age, population, and geographic characteristics—contribute to the final value.

2. Linear Regression Leaner

The Linear Regression Learner is a supervised learning algorithm that models the relationship between a set of input features and a continuous target variable by fitting a straight line (or hyperplane) through the data. When applied to predicting median house value, the Linear Regression Learner provides a clear understanding of how each feature—such as median income, housing age, or average number of rooms—contributes to the predicted price.

3. Random Forest

The Random Forest algorithm is an ensemble learning method that builds many decision trees and combines their predictions to achieve higher accuracy and stability. Each tree is trained on a random subset of the data and features, which reduces overfitting and improves generalization compared to a single decision tree. When predicting median house value, Random Forest is highly effective because it can capture complex, nonlinear relationships between variables such as income, location, population, and housing characteristics.

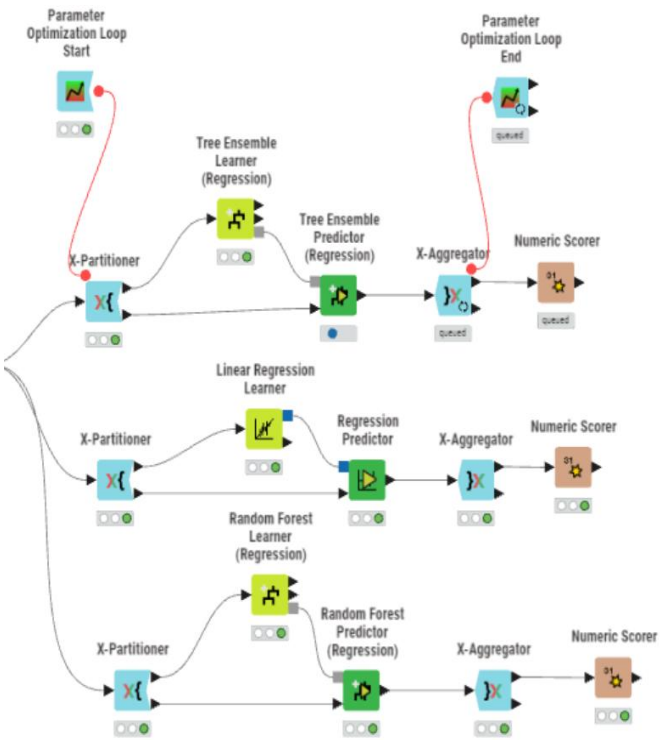

Figure 14: Machine Learning Regressors

3.2.2 Comparison

For comparing the different regressors 10-fold cross validation method is used. It evaluated the performance of machine learning models and made the efficient use of the dataset. This method splits the dataset into 10 equal parts (folds). The model is trained and evaluated 10 times, each time using a different fold as the test set and the remaining 9 folds as the training set.

The regressors are compared based on the two measures mean square error (MSE) and R^2. A lower MSE value represents the best model and its R^2 value is high. By analysing the values given in tables, we can conclude that Random Forest method is the better model among these.

Rows: 7 | Columns: 1

#	RowID	Prediction(LinearRegression) Number (Float)
1	R^2	0.222
2	mean absolute error	78,397.285
3	mean squared error	10,363,309,723.201
4	root mean squared error	101,800.342
5	mean signed difference	3,481.438
6	mean absolute percentage error	0.472
7	adjusted R^2	0.222

Figure 15: Prediction (Linear Regression) Table

Rows: 7 | Columns: 1

#	RowID	Prediction (RandomForest) Number (Float)
1	R^2	0.824
2	mean absolute error	32,283.311
3	mean squared error	2,339,059,548.194
4	root mean squared error	48,363.825
5	mean signed difference	141.897
6	mean absolute percentage error	0.183
7	adjusted R^2	0.824

Figure 15: Prediction (Random Forest) Table

Rows: 7 | Columns: 1

#	RowID	Prediction (Tree Ensemble Predictor) Number (Float)
1	R^2	0.824
2	mean absolute error	32,332.405
3	mean squared error	2,343,644,297.958
4	root mean squared error	48,411.2
5	mean signed difference	123.976
6	mean absolute percentage error	0.184
7	adjusted R^2	0.824

Figure 16: Prediction (Tree Ensemble Predictor) Table

3.3 Model Optimization

To optimize the Random Forest Regression model, GridSearchCV is used to extensively tune the number of estimators or trees (n_estimators). This leads to an enhanced accuracy of a more generalized model, as it tries to find better prediction performance on unseen data by doing a systematic analysis of the given predefined parameter grid that covers values such as 300 for n_estimators.

3.4 Train and Save the Model

Similar to Classification, the best model is now trained on the whole dataset and then is saved so that it can be used for further reference.

Conclusion

In conclusion, the "Adult Income" and "Housing" datasets were used effectively in this investigation to do classification and regression utilizing machine learning techniques. Following careful data pre-processing, exploratory data analysis, and model selection, Random Forest was determined to be the most appropriate model for both tasks, yielding the greatest combination of accuracy and low error rates. With 83.2% classification accuracy and 0.82 R^2 for regression, hyperparameter tuning also significantly improved model performance. Therefore, the findings highlight the significance of model optimization and the broad applicability of Random Forest to enormous data, as is the case in practical predictive modelling.